

NAME: THULASI.S

DATE:21.02.26

ROLL NO: 241901118

EXPERIMENT 6

DEMONSTRATE LINUX PRIVILEGE ESCALATION

AIM:

To demonstrate and understand Linux privilege escalation techniques using the TryHackMe , and to identify common misconfigurations that allow a normal user to gain elevated (root) privileges.

ALGORITHM:

1. Connect to the TryHackMe virtual machine using SSH or the provided web terminal.
2. Connect to the TryHackMe virtual machine using SSH or the provided web terminal.
3. Search for vulnerabilities using tools like find, linpeas or manual checks.
4. Use the discovered vulnerability to escalate privileges.
5. Confirm root access using whoami or id, ensuring the user has successfully gained elevated privileges.

TASKS:

[← Back to all walkthroughs](#)



Linux Privilege Escalation

Learn the fundamentals of Linux privilege escalation. From enumeration to exploitation, get hands-on with over 8 different privilege escalation techniques.

📶 50 min 👤 161,974 🏆 6

[Share your achievement](#) [Start AttackBox](#) [Save Room](#) [5884 Recommend](#) [Options](#)

Room completed (100%)

- Task 1 Introduction
- Task 2 What is Privilege Escalation?
- Task 3 Enumeration
- Task 4 Automated Enumeration Tools
- Task 5 Privilege Escalation: Kernel Exploits
- Task 6 Privilege Escalation: Sudo
- Task 7 Privilege Escalation: SUID
- Task 8 Privilege Escalation: Capabilities
- Task 9 Privilege Escalation: Cron Jobs
- Task 10 Privilege Escalation: PATH
- Task 11 Privilege Escalation: NFS
- Task 12 Capstone Challenge



Linux Privilege Escalation

Learn the fundamentals of Linux privilege escalation. From enumeration to exploitation, get hands-on with over 8 different privilege escalation techniques.

📶 50 min 👤 161,974 🏆 6

[Share your achievement](#) [Start AttackBox](#) [Save Room](#) [5884 Recommend](#) [Options](#)

Room completed (100%)

Task 1 Introduction

Privilege escalation is a journey. There are no silver bullets, and much depends on the specific configuration of the target system. The kernel version, installed applications, supported programming languages, other users' passwords are a few key elements that will affect your road to the root shell.

This room was designed to cover the main privilege escalation vectors and give you a better understanding of the process. This new skill will be an essential part of your arsenal whether you are participating in CTFs, taking certification exams, or working as a penetration tester.

Answer the questions below

Read the above.

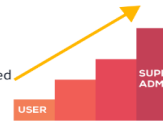
No answer needed

✓ Correct Answer

Task 2 What is Privilege Escalation?

What does "privilege escalation" mean?

At its core, Privilege Escalation usually involves going from a lower permission account to a higher permission one. More technically, it's the exploitation of a vulnerability, design flaw, or configuration oversight in an operating system or application to gain unauthorized access to resources that are usually restricted from the users.



Why is it important?

It's rare when performing a real-world penetration test to be able to gain a foothold (initial access) that gives you direct administrative access. Privilege escalation is crucial because it allows you to gain system administrator levels of access, which allows you to perform actions such as:

- Resetting passwords
- Bypassing access controls to compromise protected data
- Editing software configurations
- Enabling persistence
- Changing the privilege of existing (or new) users
- Execute any administrative command

Answer the questions below

Read the above.

No answer needed

✓ Correct Answer

Answer the questions below

What is the hostname of the target system?

wade7363

✓ Correct Answer

What is the Linux kernel version of the target system?

3.13.0-24-generic

✓ Correct Answer

What Linux is this?

Ubuntu 14.04 LTS

✓ Correct Answer

What version of the Python language is installed on the system?

2.7.6

✓ Correct Answer

What vulnerability seem to affect the kernel of the target system? (Enter a CVE number)

CVE-2015-1328

✓ Correct Answer

Answer the questions below

find and use the appropriate kernel exploit to gain root privileges on the target system.

No answer needed

✓ Correct Answer



What is the content of the flag1.txt file?

THM-28392872729920

✓ Correct Answer

How many programs can the user "karen" run on the target system with sudo rights?

3

✓ Correct Answer

What is the content of the flag2.txt file?

THM-402028394

✓ Correct Answer

How would you use Nmap to spawn a root shell if your user had sudo rights on nmap?

sudo nmap --interactive

✓ Correct Answer

What is the hash of frank's password?

\$6\$2.sUUDsOLIpXKxcr\$elmtgFExyr2ls4jsghdD3DHLHHP9X50lv.jNmwo/BJpphrPRJWjelWEz2HH.joV14aDEwW1c3CahzB1uaq

✓ Correct Answer

Answer the questions below

Which user shares the name of a great comic book writer?

✓ Correct Answer

What is the password of user2?

✓ Correct Answer

What is the content of the flag3.txt file?

✓ Correct Answer

Answer the questions below

Complete the task described above on the target system

✓ Correct Answer

How many binaries have set capabilities?

✓ Correct Answer

What other binary can be used through its capabilities?

✓ Correct Answer

What is the content of the flag4.txt file?

✓ Correct Answer

How many user-defined cron jobs can you see on the target system?

✓ Correct Answer

What is the content of the flag5.txt file?

✓ Correct Answer

What is Matt's password?

✓ Correct Answer

Answer the questions below

What is the odd folder you have write access for?

✓ Correct Answer



Exploit the \$PATH vulnerability to read the content of the flag6.txt file.

✓ Correct Answer



What is the content of the flag6.txt file?

✓ Correct Answer

How many mountable shares can you identify on the target system?

✓ Correct Answer

How many shares have the "no_root_squash" option enabled?

✓ Correct Answer

Gain a root shell on the target system

✓ Correct Answer

What is the content of the flag1.txt file?

✓ Correct Answer

What is the content of the flag2.txt file?

✓ Correct Answer

RESULT:

All the tasks in Linux Privilege Escalation tryhackme were completed successfully.